

Rufat Asadli

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EDUCATION

- **ETH Zurich [5.5/6.0]** Sep'23-Oct'25
MSc in Statistics
- **Charles University [1.1/1.0]** Sep'19-Jun'22
BSc in Economics and Finance
 - Recipient of **Merit Stipend for Outstanding Academic Achievements** for 2 years awarded to **top 10%** students across all departments.

EXPERIENCE

- **Networked Systems Group, ETH Zurich** Oct'25 - Present
Research Intern, Supervised by Prof. Dr. Laurent Vanbever Zurich, Switzerland
 - Extending previous research on **benchmarking** LLMs for **code generation** and network **configuration synthesis** with novel **curriculum learning** algorithms.
- **NetFabric** Mar'25 - Oct'25
Graduate Student Researcher, Supervised by Prof. Dr. Laurent Vanbever and Dr. Benjamin Bichsel Zurich, Switzerland
 - Developed the first robust **evaluation benchmark** of closed- and open-source LLMs for **constrained code generation** tasks via zero-shot learning and supervised fine-tuning experiments.
- **Rycolab, ETH Zurich** Apr'24 - May'25
Graduate Student Researcher, Supervised by Prof. Dr. Ryan Cotterell and Dr. Alex Warstadt Zurich, Switzerland
 - Developed a PPO-based training setup where a **speaker language model (LM)** learns to communicate privileged info to a **listener LM**, receiving rewards based on the listener's comprehension accuracy.
- **Kapital Bank OJSC** Oct'22 - Sep'23
Data Science Researcher, AI Center of Excellence Baku, Azerbaijan
 - Built a **high-frequency prediction model** using **ensemble methods** together with Fourier transforms for feature engineering, **calibrating** raw predictions with Monte Carlo sampling from error distribution. Affiliated as a student researcher after Sep' 23 remotely.

PROJECTS

- **Towards Developmentally Plausible Rewards: Communicative Success as a Learning Signal for Interactive Language Models** [\[Code\]](#) [\[Paper\]](#) Apr'24-May'25
Preprint
 - We present an **interactive language model training** method inspired by child language acquisition, showing communication-based rewards signal grammaticality and shape speaker behavior.
- **Does Catastrophic Forgetting Happen in Tiny Subspaces?** [\[Code\]](#) [\[Report\]](#) Oct'24-Jan'25
For Deep Learning, ETH Zürich. Grade: 6/6
 - Explored **neural network subspace structure in continual learning** showing that task-specific model learning and forgetting primarily occurs in gradient subspaces with small eigenvalues.
- **TARMAC: Conversational AI Dispatcher** [\[Website\]](#) Jul'24 - Dec'24
Startup Project
 - Contributed to a **conversational dispatch system** with private data **fine-tuning** and dynamic **chat buffer memory update**, learning to **convert user prompts into SQL queries** for truck data retrieval.

SKILLS

- **Programming Languages:** Python, HTML, CSS
- **Packages and Softwares:** PyTorch, TensorFlow, Transformers, NLTK, LangChain, TRL, Unsloth, vLLM, Accelerate, NumPy, Git, Bash, Docker, SQL, L^AT_EX

AWARDS AND ACHIEVEMENTS

- Recipient of State Program scholarship to study Master's degree abroad (Azerbaijan) 2023
- [TensorFlow Developer Certificate](#) 2022